INDIAN RAILWAYS & SAFETY TELEMETRY BOARD

STANDARD OPERATING PROCEDURE (SOP) & TECHNICAL MANUAL:
AI Railway Collision Prevention, Kavach ATP System & Wildlife Safety Directives
Doc Reference: IR-SOP-KAVACH-WILDLIFE-2024-V4 | SIL-4 Certified Specification

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1. Executive Summary & Kavach ATP Architecture

Kavach is India's National Automatic Train Protection (ATP) system, developed by the Research Designs and Standards Organisation (RDSO) in association with Indian industry. Kavach operates under Safety Integrity Level 4 (SIL-4) standards to prevent collisions, enforce speed compliance, and eliminate Signal Passing at Danger (SPAD).

1.1 Core Hardware and Subsystems:

- **Locomotive Kavach Unit (Loco-Kavach):** On-board computer unit housed in the engine cab, interfaced with Driver Machine Interface (DMI), Brake Interface Unit (BIU), UHF radio transceiver (406-470 MHz), and under-chassis RFID reader antenna.
- **Station Kavach Unit:** High-availability dual-redundant processing unit deployed at railway stations, connected to the Electronic Interlocking (EI) and optical track telemetry relays.
- **Trackside RFID Network:** Passive RFID tags clamped onto sleepers along the track center. Spacing is maintained at 1000m on open tracks, 100m approaching signals, and 50m in turnout/loop lines to determine precise train location, heading, and track ID.
- **UHF Radio & GSM-R/LTE-R Mesh:** Continuous bi-directional wireless telemetry between trains, stations, and level crossing gates with transmission cycle time under 500 milliseconds.

2. Collision Prevention Protocols & Automatic Braking Curves

Kavach continuously calculates dynamic target distance curves based on train weight, locomotive brake capacity, gradient profile, and track friction coefficients.

- **Head-on Collision Prevention:** When two locomotives on the same track detect conflicting Movement Authorities (MA) within a 3.0 km envelope, Loco-Kavach units exchange peer-to-peer UHF collision packets and instantly trigger the Emergency Brake (EB) on both consists without requiring driver intervention.
- **Rear-end Collision Mitigation:** Loco-Kavach calculates relative separation distance using Station Kavach block occupancy telemetry. If separation drops below the Safe Braking Distance (SBD), Service Brake (SB) is engaged; if separation drops under 600m, Emergency Brake (EB) brings the following train to an absolute halt.
- **SPAD (Signal Passed at Danger) Elimination:** If a locomotive crosses an approach signal at RED (Stop aspect), Loco-Kavach triggers full emergency brake deceleration (-0.85 m/s^2), sounding cab sirens and transmitting automated broadcast alerts to Section Control.
- **Manual & Remote SOS Function:** Drivers can activate the red Mushroom SOS button on the DMI. Additionally, Station Masters or AI track sensors can inject a digital Remote SOS command that immediately stops all locomotives within a 3000-meter radius.

3. Track Telemetry & Continuous Sensing Infrastructure

Modern track safety incorporates multi-modal telemetry sensors along vulnerable track corridors to capture real-time physical conditions:

- **High-Frequency Axle Counters (HASSDAC):** Solid-state electromagnetic wheel detection coils measuring axle counts and wheel flange direction at block boundaries to ensure zero undetected decoupled rolling stock.

- **Distributed Acoustic Sensing (DAS):** Fiber-optic cables buried adjacent to rails function as distributed microphones. Acoustic vibrations caused by boulder falls, landslides, or animal footfalls within 10 meters of the track trigger instant alerts to telemetry servers.
- **Rail Temperature & Rail Buckle Telemetry:** Remote IoT thermal sensors measure rail temperature. If rail temperature exceeds 60°C in peak summer (Tmax + 25°C), speed restrictions (Caution Orders) of 30 km/h are enforced automatically to prevent track buckling.

4. Wildlife and Animal Collision Prevention Protocols (Focus: Elephant Corridors)

Elephants and other large wildlife frequently traverse recognized corridors bisecting railway tracks in the Western Ghats (Walayar, Coimbatore-Palakkad), Northeast Frontier (Alipurduar, Lumding), and Odisha forest tracks. Indian Railways mandates the following AI and optical intervention layers:

- **AI Optical & Thermal Early Warning Towers:** Thermal and optical camera towers placed every 400m in designated wildlife corridors stream video to AI edge processors. The system recognizes elephants, cattle, leopards, and human trespassers at distances up to 800 meters during day, night, fog, and heavy rain.
- **Automated Speed Restriction (ASR) Zones:** Between 18:00 hrs and 06:00 hrs, trains traversing elephant corridors are speed-restricted to a maximum of 30 km/h to guarantee a safe braking stopping distance.
- **Plan Bee Acoustic Deterrent System:** Specially engineered directional acoustic devices broadcast recorded sound of swarming honey bees (proven natural deterrent for Asian elephants) up to 600 meters along tracks, clearing herds without causing panic.
- **Intrusion Detection System (IDS) Footstep Classification:** Seismic and fiber optic ground sensors detect heavy animal footsteps (frequency signature 10 Hz - 25 Hz) within 15 meters of the ballast. Upon 3 consecutive detected paces, a Tier-1 Critical Hazard is raised.

5. Emergency Action Protocols & Threat Severity Levels

Alert Level	Trigger Event	Kavach Automated Action	Loco Pilot Directive
CRITICAL (RED)	Elephant / Cattle / Vehicle / Boulder detected within 1,000 m	Automatic Emergency Brake (EB) application	Confirm EB application, halt train
HIGH (ORANGE)	Wildlife herd spotted within 50 m of track	Automatic Speed Restriction (ASR) engaged	Reduce speed to 20 km/h, activate Plan B
MEDIUM (YELLOW)	Nighttime corridor entry / Track anomaly detected	Advisory caution flag displayed on DMI	Maintain vigilance, reduce speed, monitor
NORMAL (GREEN)	Clear track ahead, valid Movement Authority	Normal operation, speed limited by Kavach	Maintain scheduled speed, observe signals

6. Bilingual Operational Directives (English & Tamil / ■■■■■)

For Southern Railway and South Western Railway zones passing through Tamil Nadu and Kerala borders, telemetry systems and voice advisories must support standard bilingual terminology:

- **Hazard Detected:** English: 'Danger! Obstacle or animal detected on track ahead.' | Tamil: '■■■■■■■■■■■! ■■■■■■■■■■■■■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■■■■■■■■■■. ■■■■■■ ■■■■■■■■■ ■■■■■■■■■■■.'
- **Elephant on Track:** English: 'Emergency! Elephant herd detected on track near forest corridor. Emergency braking engaged.' | Tamil: '■■■■■ ■■■■■■■■■■■! ■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■ ■■■■■ ■■■■■■■. ■■■■■■■■ ■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■.'
- **Speed Restriction:** English: 'Caution! Elephant corridor active. Speed restricted to 20 km/h.' | Tamil: '■■■■■■! ■■■■■■■■ ■■■■■■■■■ ■■■■■■. ■■■■■■ 20 ■■.■■ ■■ ■■■■■■■■■■■■■■■■■■■■■■.'

- **Track Clear:** English: 'Track is clear. All telemetry normal. Safe to proceed.' | Tamil: '[illegible]. [illegible].'

Approved by Railway Safety Commission & Wildlife Protection Division. Document version 4.2. For automated AI query verification and telemetry cross-reference.